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ORIGINAL ARTICLE

A classification study for Turkish folk music makam recognition using machine learning with data augmentation techniques

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Abstract

Makam is defined as melodies that are described with typical the agâz(beginning), seyir (the orientation style), and karar (ending) features in a certain perde düzen (tone/fret tunings). For this reason, determining the makam is the basic step in understanding the melodic progression, as in musical keys. Machine learning, an artificial intelligence discipline, offers stable solutions for estimating different situations based on known data. In this study, classifications for makam recognition in Turkish folk music were carried out using different machine learning methods on the original dataset. The success of the different methods used was compared, and also data augmentation was performed by using SMOTE, KMeansSMOTE, ADASYN, and SVMSMOTE oversampling techniques, which are widely used in the literature to increase the prediction success of the classifiers. For each of eight different machine learning techniques, namely light gradient boosting machine, eXtreme gradient boosting, naive Bayes, decision tree, support vector machine, K-nearest neighbor, random forest, and logistic regression, in addition to classification without data augmentation, a total of 40 classification processes were carried out using four different data augmentation techniques. The results of the classification processes are reported with four different parameters: accuracy, precision, recall, and F1-score. The highest accuracy value of 99.17% was obtained when the light gradient boosting machine classifier was used together with the SMOTE data augmentation technique. It was observed that the measurements obtained in the study were higher than the results obtained in similar studies conducted in the literature in recent years.

Keywords Turkish folk music · Makam recognition · Machine learning · Oversampling

1 Introduction

The structures that provide the melody design in the traditional music of Anatolia and the surrounding geographies are expressed with the concept of "makam." As such, the makam may overlap with the concept of tonal or modal scale in western music. But it has some differences from it. Unlike the scale structure, makam is also a structural model in which the melodic progression is expressed. In this context, a makam can be viewed as a musical setting with (a) a well-defined underlying scale of pitches and (b) a set of conventions that structure the temporal ordering of the pitches into melodic lines [1, 2]. A glance at the temporal definitions will give a deeper understanding of what makam is. These are as follows:

The multifaceted essence of the concept of "makam" is comprehensively expounded upon by an array of scholarly elucidations. Noteworthy instances of these elucidations include the perspectives of Yekta[2], Arel[3], Karadeniz[4], Gürmeriç[5], and Stubbs[6], which collectively contribute to a deeper comprehension of this intricate musical phenomenon. Yekta's analysis from 1923 introduces the notion that "makam" embodies a process of configuration, wherein it materializes as a distinct manifestation within the musical scale paradigm. This manifestation, achieved through the intricate arrangement of

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intervals and complex relational structures, signifies a nuanced orchestration of tonal components. Similarly, Arel's examination in 1968 probes into the intricate interplay between pitches inherent within a given scale or melody. The interaction of these pitches, notably in relation to the tonic and/or dominant notes, gives rise to the unique tonal quality termed "makam." This perspective emphasizes the dynamic relationship among tonal elements. Karadeniz's delineation in 1980 further enriches the discourse by characterizing "makam" as the very essence engendered by melodic phrases governed by a set of progression rules, denoted as "melodic progressions." These progressions are confined within the context of a specific scale, thus imbuing each "makam" with its distinct musical identity. In 1966, Gürmeriç underscores the foundational role of the scale within the "makam" framework. He accentuates the evolutionary nature of "makam," conceptualizing it as a progression that breathes vitality into its tonal identity. The process initiates from a specific point within the sequence and progressively unfolds, culminating in intensified potency and eventual resolution. Taking a pragmatic stance, Stubbs' perspective from 1994 introduces the concept of "makam" as a functional theory of melody. This approach categorizes melodies into discrete families or categories, distinguished by meticulous microtonal adjustments applied to specific tones, as guided by tradition. These nuanced variations harmonize with idealized contours of melodic structure, enriching the sonic tapestry. It is vital to recognize that, akin to various intricate musical concepts, terse definitions or explanations only scratch the surface. A more profound grasp of "makam" and its intricate web of interpretations can be attained through a more comprehensive exploration, notably through the scholarly works of Can[7] and Ayangil[8]. These scholars offer a deeper analytical voyage, unveiling the intricate layers and implications of "makam" within the realm of musical theory and practice. Therefore, the culmination of these multifarious perspectives leads to a more holistic and insightful understanding of the profound essence encapsulated within the concept of makam. Today, the concept of makam has started to be defined in a way in which melodic progression and melodic structures are at the forefront. In this framework, with a general and understandable definition, makam; agâz/beginning, seyir/orientation, and karar/ending characteristics are typical tunes in a certain tone/fret tunings [9]. The makam theory describes how the tuning system, which can be seen concretely in instruments, is designed to produce melody through written and verbal rules [10]. Based on these definitions; In the makam classification and the makam/terkib(compound makam structure) distinction, first of all, the characteristics such as the tone/fret tunings in which the melody takes place, the âgaz pitch or the central pitch of the âgâz melody, the mode of seyir, if any, the temporary ending and the ended pitch should be taken into consideration [11]. The melody formed with these qualities cannot be divided into other parts by carrying a unique integrity and will be separated in a way that leaves no room for doubt in other makams [11]. Makam and Usûl (rhythm) constitute the most basic points of a Turkish music piece. The most basic step in the analysis of Turkish music is the makam classification. This situation constitutes the beginning of the other analysis steps (form, style, region, etc.) that will follow.

With the emergence of the computational musicology discipline recently, as in almost all world music [12] many researchers have started to examine Turkish music using various mathematical methods [1, 6–13]. In such studies, it is seen that easy-to-access, categorized data stacks are extremely important for researchers. Researchers today believe that music; whether intelligence and creativity meet, embellished with emotional interactions and occur randomly; the degree to which the structure of the works produced mathematically follows a certain system; it seeks various answers in line with the possibilities provided by today's technology on issues such as how and according to what the relativity criteria of the feeling called admiration change [21]. Since the studies are usually related to large databases, computers with complex algorithms have become an integral part of this discipline today. Thanks to today's computers with high processing speeds, time-consuming intensive operations such as counting, sorting, grouping, finding patterns, deleting, adding, etc. in large datasets in databases can be performed in a very short time. Various file types are used to perform these operations. Thanks to the definition and adoption of formats such as Humdrum [22], MIDI, MusicXML [23] and SymbTr [24], these processes can be done more easily and data exchange between researchers can be provided easily. The use of such opportunities in music research is important for the preservation, protection, more accurate evaluation, and obtaining definitive results of musical information, which is based on personal opinions and experiences, transmitted from person to person in this field. In this field, Music Information Retrieval (MIR) is a rapidly developing field that has gained importance recently due to the widespread use of databases containing sound recordings and symbolic data, and the continuous increase in access to information over the internet. Methods in this field, most of which are based on Western music processing, are based on Western music theory, and a significant number of studies use only symbolic data encoded with Western music notation [16].

Today, the Turkish music dimension of MIR has been significantly improved thanks to the analysis tools [7, 8, 12] and corpus developed as a result of various researches on the analysis and detection of Turkish music.

However, the dimension of Turkish Folk Music (TFM) has not been sufficiently examined and analyzed in this sense. It can be said that this situation is caused by the fact that different [18–21] and few opinions were put forward to address and explain the field of folk music from a “theoretical” perspective. As a matter of fact, the concept of authority so far; melodic progression style, melody pattern, melodic nuclei, scale, pattern motifs, pitch hierarchy, interval order etc. addressed from various perspectives [21–23]. TFM’s melodic structure and sound system and Ottoman/Turkish music (OTM) contain some theoretical differences. Basically, the pitch differences between the TFM and the Arel-Ezgi-Uzdilek (AEU) system [31], which is widely used in OTM, are presented in Table 1.

As seen in Table 1, while the TFM fret system is performed with a system consisting of 17 pitches per octave, a fret system with 24 pitches per octave is used in OTM. Although the pitch differences between these systems have some differences in terms of ratio and logarithmic cents, the melodic progression (seyir) and pattern relations show great similarities. As a matter of fact, the two genres in question are not separated from each other with sharp lines but appear as the reflection of two different styles connected to the same root [25, 26]. In OTM, artistic creation and melodic production based on a rich compositional seyir cause the makams to be perceived more mixed, and the use of personal or school-based pitch-motif to be perceived as a makam rule. In this case, problems will arise if every difference in the makam descriptions and the composer’s ornaments or uses contrary to the makam in every piece are perceived as a feature of the makam [11]. Folk music; it provides a wide range of data that can be studied theoretically for researchers, since it contains examples that fit perfectly with the definitions and descriptions of makams with its simple melodic structure. It should be remembered that the development of makam theory would not have been possible without the existing repertoires. Since the folk music repertoire is obtained through compilations, it is inevitable to use a historical perspective in the analysis and classification of the pitches and melodic progression styles encountered. Because the melody repertoires, on the one hand, ensure the transmission of certain works and typical motif styles identified with these works, on the one hand, and on the other hand, they contain perfect examples in

Table 1 Comparison of TFM and OTM frets with the base pitch in terms of ratio and logarithmic cents

Step	OTM fret system (AEU)			TFM fret system			Difference
	Pitch	Ratio	Cent	Pitch	Ratio	Cent	
0	Kaba Çargâh	1/1	0.00	La	1/1	0	0
1	Kaba Nim Hicaz	256/243	90	–	–	–	–
2	Kaba Hicaz	2187/2048	114	Sib	18/17	99	−15
3	Kaba Dik Hicaz	65,536/59049	180	Sib2	12/11	151	−29
4	Yegâh	9/8	204	Si	9/8	204	0
5	Kaba Nim Hisar	32/27	294	–	–	–	–
6	Kaba Hisar	19,683/16384	318	Do	81/68	303	−15
7	Kaba Dik Hisar	8192/6561	384	Do#3	27/22	355	−29
8	Hüseyniaşiran	81/64	408	Do#	81/64	408	0
9	Acemaşiran	4/3	498	–	–	–	–
10	Dik Acemaşiran	177,147/131072	522	Re	4/3	498	-24
11	Irak	1024/729	588	–	–	–	–
12	Geveşt	729/512	612	Mib	24/17	597	−15
13	Dik Geveşt	262,144/177147	679	Mib2	16/11	649	−30
14	Rast	3/2	702	Mi	3/2	702	0
15	Nim Zirgüle	128/81	792	–	–	–	–
16	Zirgüle	6561/4096	816	Fa	27/17	801	−15
17	Dik Zirgüle	32,768/19683	882	Fa#3	18/11	853	−29
18	Dügah	27/16	906	Fa#	27/16	906	0
19	Kürdi	16/9	996	–	–	–	–
20	Dik Kürdi	59,049/32768	1020	Sol	16/9	996	−24
21	Segâh	4096/2187	1086	–	–	–	–
22	Buselik	243/128	1110	Lab	32/17	1095	−15
23	Dik Buselik	1,048,576/231441	1177	Lab2	64/33	1147	−30
24	Çargâh	2/1	1200	La	2/1	1200	0

terms of reflecting the changes that occur on the tunes through various innovations and interaction [9]. In this framework, the concept of makam constitutes the most basic structure in the theory, production and transmission dimension of folk music. In the correct transmission of the melody produced, the correct expression of the makam both cognitively and practically constitutes an inevitable situation for the transfer of musical competence and theoretical phenomenon. In a framework where folk melodies are put forward irtically without any concern for producing any work of art, it is considered important to know and transfer the concept of makam correctly in order to preserve the intangible cultural heritage correctly. Because makam is a phenomenon that forms the main body of the melodic structure. Any deterioration or change that may occur in the makam structure may cause the deterioration of the entire melodic context inductively.

Especially, in TFM makam classification, various definition difficulties can be experienced with the effect of the melodic progression. It is thought that the analysis of the folk music repertoire, which contains the makam structures in all its simplicity and purity, with this type of computer-aided analysis will make very important contributions to the theoretical field [21, 27]. Computer technology and support systems are needed to quickly resolve this phenomenon and reveal different perspectives. In recent years, developments in artificial intelligence technology also offer alternative solutions in the field of computational musicology. Machine learning, which enables to produce reliable predictions about various makams, genres, composers, instruments, etc. by learning from existing data, is a technology that has been rapidly increasing in popularity recently. In machine learning, stable predictions about new situations can be produced based on the parameters that affect a certain result, based on the available data.

The computational approach to Turkish music has been adopted in order to provide useful tools in areas such as learning, teaching, comparing this music with other music, and accessing music [35]. Expressing Turkish music both visually and audibly on the computer was with Mus2-Alfa, the first attempts were made in 1988 [36]. In the continuation of this, the designed SymbTr corpus [24] can be shown as the first comprehensive database in this field. In the literature, there are studies carried out using various statistical data analysis and machine learning techniques on sound recordings and different datasets, especially the SymbTr collection, for recognizing makams in Turkish music. Various classification algorithms have been used in these studies.

Alpkoçak and Gedik [37] using MIDI files, achieved 97.70% success in recognizing the makam using the N-gram method over 10 makams and 200 piece. Gedik and Bozkurt [38], using various Synthetic and real audio files, achieved 92% success in recognizing the makam with the Pitch interval histogram on 9 makams and 118 records. Again Gedik and Bozkurt [39] using various sound recordings, achieved a success rate of 77.38% with a Pitch frequency histogram-based method on 9 makams and 172 tracks. Ioannaidis et al. [40], using various sound recordings, achieved 74.14% success with Chroma Features on 9 makams and 302 pieces. Kalender et al. [41], achieved a 95.83% success rate with Combined Neural Networks on 9 makams and 50 pieces. Kalaycı and Kuruoğlu [42], achieved 70% success in recognizing makams with K-means and ANN algorithms on 7 makams and 103 songs using the Bir Şarkıdır Yaşamak- Turkish Art Music Selection Series dataset. Ünal et al. [20], using the SymbTr collection, achieved success in recognizing 87.90% of makams with an N-gram-based hierarchical approach on 13 makams and 877 pieces. Kızrak et al. [43], achieved 89.40% success in recognizing makams with Probabilistic Neural Networks on 6 makams and 62 pieces using various sound recordings. Kızrak and Bolat [44], achieved 92.70% success with Deep Belief Networks on 6 makams and 62 pieces using various sound recordings. Again, Kızrak and Bolat [45], using various sound recordings, achieved 96.57% success in recognizing the makam on 7 makams and 93 tracks with Deep Belief Networks. Abidin et al. [13], again using the SymbTr collection, achieved a success rate of 89.70% with decision trees on 13 makams and 1261 pieces. Öztürk et al. [18], again using the SymbTr collection, achieved success in recognizing 88.12% of makams with various classifiers on 13 makams and 1246 pieces. Demirel et al. [46], achieved 89% success with Chroma Features on 20 makams and 997 pieces using OTM recognition dataset. Kayış et al. [47], using various audio files, achieved 98.21% success in recognizing makams with SVM on 8 makams and 448 tracks. Börekci [48], using a dataset created from the TFM repertoire, obtained the highest accuracy of 90.15% with the XGBoost algorithm in his study using various machine learning techniques on 8 makams and 437 pieces.

In this study, a high-performance solution for makam recognition in Turkish folk music is presented by using different machine learning methods on the original dataset. The successes of the different methods used were compared and also different data augmentation techniques were applied to increase the current prediction success of the classifiers. The results of the classification processes were reported with 4 different parameters, namely accuracy, precision, recall and F1-score, and the results of 40 measurements were interpreted.

2 Material and metod

In this study, classifications were carried out using different machine learning techniques to determine the makam belonging of Turkish folk music works. For this purpose, a dataset containing 28 features and 504 samples was used. In order to determine the makam belonging of Turkish folk music works, classifications were carried out using different machine learning techniques. For classification, light gradient boosting machine, eXtreme gradient boosting, naive Bayes, decision trees, support vector machines, K-nearest neighbor, random forest, logistic regression methods were used and their performances were evaluated on different metrics. In this direction, in addition to the default success of eight different methods used in the study, four different data augmentation methods were applied independently for each, and the successes of forty separate classification processes were reported.

2.1 Dataset

In this study, which aims to provide a solution for recognizing makams on Turkish folk music works, a new, updated and expanded version of the dataset previously created by Börekci [48], which was restructured for this study, was used. The artifacts in the original dataset were obtained from a public data repository, which is accepted as a reliable source for studies in the field and used in many studies [25, 49, 49, 50], [51]. The artifacts in this dataset consist of notation and metadata dictated by a group of expert musicians of folk melodies collected in Anatolia over a historically long period of time. Within the scope of this study, 437 sample artifacts consisting of 21 attributes in the existing dataset were updated in the form of 504 sample artifacts consisting of 28 attributes and a unique new dataset was created.

2.1.1 Creating the dataset

The data were first transferred to the computer with the Finale 2014 notation program and recorded here in MusicXML format. A database was created by converting the works prepared in MusicXML format into SymbTr text format, which is designed to represent Turkish music in a complete way, with the help of a converter. Examples of the SymbTr format are presented in Fig. 1 and Table 2.

On the created dataset, the attribute catalog in jSymbolic [52] and the parameters that were found to be important in recognizing the makam [2, 43, 44] were determined and feature inferences were made. The makam labeling of the pieces was made by three experts with reference to Öztürk's [9] approach and was cross-validated. In the data of each piece in the dataset, there are 28 input features and one makam class label. Target class labels have eleven different values within the framework of the makam to which each pieces belongs. In the dataset, there are 504 works in total, including 111 Hüseyni, 83 Uşşak (Dügâh-ı Kadim), 77 Hicaz, 77 Nevruz-ı Asil, 39 Uzzal, 27 Rast, 23 Gülizar, 19 Muhayyer, 19 Nühüft-i Kadim, 16 Nikriz (Niyriz) and 13 Muhayyer-Kurdi. The melody styles, which are described as two different makams with the names Nikriz-i Sagir and Nikriz-i Kebir, according to the pitch mentioned in Öztürk's approach, were combined under the same makam title in this study and named as Nikriz due to the small number of samples. Details on the characteristics of the dataset are presented in Table 3. The statistical characteristics of the features in the dataset are given in Table 4.

Correlations between the features of the dataset are given in Fig. 2. On the color scale, light colors indicate high correlation and dark colors indicate low correlation.

When the correlation matrix is examined, although there are features that have a high correlation with each other, there is no feature that has a correlation of more than 0.5 with the target class. This situation shows that there is no dominant characteristic in predicting the target class.

2.2 Machine learning algorithms used for classification

In this study, light gradient boosting machine, eXtreme gradient boosting, naive Bayes, decision trees, support vector machines, K-nearest neighbor, random forest, logistic regression classification algorithms, which are frequently preferred in machine learning-based researches, were used.

Light gradient boosting machine (LGBM) is an effective gradient boosting decision tree application developed by Microsoft with gradient-based one-side sampling and exclusive feature bundling to increase computational efficiency without harming accuracy. In this context, it is mainly based on decision tree algorithms and is used for sorting, classification, and other machine models. LGBM is an ensemble model whose base classifier is decision tree, which could be trained in sequence by fitting the negative gradients of loss function [55]. LGBM has the potential to provide high performance in datasets with complex relationships due to its advantages such as fast prediction capability, good adaptation to imbalanced class problems, and high prediction performance.

eXtreme gradient boosting (XGB), originally developed by Friedman in 2002, is a machine learning technique based on gradient boosting and decision tree algorithms [56] and works with the logic of turning individual weak learners into strong learners. In the objective function of

Fig. 1 Score notation of a music piece in Turkish folk music repertoire

Table 2 SymbTr representation of the notes in Fig. 1

Code	Note 53	Comma 53	Note AE	Comma AE	Num	Denom	MS	LNS	VelOn	Syll
9	Re6	D6	380	380	1	32	125	95	96	A
9	Re6	D6	380	380	3	32	375	95	96	MAN
9	Re6	D6	380	380	1	32	125	95	96	CE
9	Re6	D6	380	380	3	32	375	95	96	RİT
9	Re6	D6	380	380	3	32	375	95	96	IRAK
9	Re6	D6	380	380	1	32	125	95	96	KA
9	Re6	D6	380	380	3	32	375	95	96	DAN
9	Re6	D6	380	380	1	32	125	95	96	DA
9	Re6	D6	380	380	1	24	167	95	96	SÖ
9	Re6	D6	380	380	1	24	167	95	96	KÜN
9	Re6	D6	380	380	1	24	167	95	96	E
9	Re6	D6	380	380	1	8	500	95	96	DİN
9	Re6	D6	380	380	3	32	375	95	96	CE
9	Mi6	E6	389	389	1	32	125	95	96	
9	Re6	D6	380	380	1	12	333	95	96	
…										

the XGB algorithm, it is aimed to find the best parameters on the given training set and to measure the model performance. XGB offers performance solutions to classification problems due to its advantages such as high prediction accuracy, fast training time and low risk of overlearning.

Bayes' theorem is an important topic studied in probability theory. This theorem shows the relationship between conditional probabilities and marginal probabilities within the probability distribution for a random variable. Naive Bayes (NB) algorithm is a classification algorithm named after mathematician Thomas Bayes. The NB classifier aims to detect the class of data whose class is unknown, with a set of calculations defined according to probability principles. The NB algorithm is a simple algorithm and is capable of fast and high accuracy estimation. The algorithm calculates the probability of each situation and classifies it according to the highest probability value. Bayes' theorem is expressed as in Eq. 1.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{1}$$

Decision trees (DT) is a tree-based algorithm that is frequently used in classification and regression problems and can work with complex datasets. DT creates a classification model by formulating a tree structure of decision nodes and leaf nodes depending on the features and target. DT is an algorithm that makes decisions by dividing the data, can adapt to class imbalance problems, reduces the need for data preprocessing, and stands out with its simple and useful structure.

The support vector machines (SVM) algorithm is a set of supervised learning methods used for classification, regression, and outlier detection [57]. Its purpose is to find a hyperplane in an N-dimensional space (N: number of features) that clearly classifies data points. In this framework, SVM creates a model that assigns new samples to one of these two classes in a non-probability manner by learning from the training dataset, each of which is marked as belonging to one of the two categories. In the plane where the data samples are located, a decision boundary is drawn at the furthest distance from the members of the two classes to separate the classes from each other. The low sensitivity of SVM to the overfitting problem and its high

Table 3 Features of the dataset

Features	Type	Definition
Target	Categorical	Makam Class (1 = Gülizar, 2 = Hicaz, 3 = Uzzal, 4 = Hüseyni, 5 = Muhayyer, 6 = Muhayyer-Kürdi, 7 = Nevruz-ı Asıl, 8 = Nikriz, 9 = Nühüft-i Kadim, 10 = Rast, 11 = Uşşak)
aOrtalama	Numeric	It expresses to the weighted MIDI pitch average calculated from the duration values of the piece's pitches
agaz	Numeric	It refers to the most frequently used pitch (peak value) in the first 10% of each piece
agaz_median	Numeric	It expresses the median MIDI pitch value of the pitches used in the first 10% of each piece
agaz_ort	Numeric	It refers to the MIDI pitch average of all pitches used in the first 10% of each piece
atepe	Numeric	This phrase refers to the peak value that can be calculated using the piece's overall weight
birinciGuclu	Numeric	It expresses to the MIDI pitch value of the piece's primary initial pitch that is utilized the most
birinciGucluSure	Numeric	It refers to the first most frequently used time value in the piece
birinciMikroGuclu	Numeric	It expresses to the MIDI pitch value of the first microtonal pitch that is played the most frequently, disregarding complete pitches (natural pitch)
dif_Rtm_Val	Numeric	It expresses to the different time values (e.g.16, 32, 64)
enKisaRtm	Numeric	It expresses to the piece's shortest time value
enPest	Numeric	It expresses to the piece's lowest pitch
enTiz	Numeric	It expresses to the piece's highest pitch
enUzunRtm	Numeric	It expresses to the longest duration value
ikinciGuclu	Numeric	It expresses to the MIDI pitch value of the piece's most utilized second pitch
ikinciGucluSure	Numeric	It refers to the second most frequently used time value in the piece
ikinciMikroGuclu	Numeric	It speaks about the second-most-used microtonal pitch in terms of MIDI pitch value, omitting complete pitches (natural pitch)
ilkNota	Numeric	It refers to the MIDI pitch value of the first (starting) pitch of the piece
notaCarpiklik	Numeric	The skewness of all the pitches used in the piece is what it alludes to
ortalama	Numeric	It refers to the MIDI pitch average of all pitches used in the piece
ortanca	Numeric	It expresses the median MIDI pitch value of the pitches used in the piece
rtmCarpiklik	Numeric	The skewness of all the timings employed in the piece is what it alludes to
rtmOrtalama	Numeric	It refers to the piece's average of the time values
sesSistemi	Numeric	By classifying the pitches that are used at least once in the work, it indicates how many pitches there are in the whole
sonNota	Numeric	It expresses to the ultimate (decision) pitch of the piece's MIDI pitch value
stdPerde	Numeric	It expresses the standard deviation of the pitches used in the piece
toplamNota	Numeric	It expresses to the overall quantity of pitches used in the piece
ucuncuGuclu	Numeric	It expresses to the MIDI pitch value of the piece's most used third pitch
varyansPerde	Numeric	It expresses the variance of codes of the pitches used in the piece

accuracy increases the prevalence of use. The effectiveness of SVM in complex datasets, its high generalization, its optimization of class distinctions, its low sensitivity to the overfitting problem and its high accuracy increase its widespread use.

The K-nearest neighbor (K-NN) algorithm, a nonparametric classification method developed by Evelyn Fix and Joseph Hodges in 1951 and later expanded by Thomas Cover, is a simple supervised machine learning algorithm that can be used to solve both classification and regression problems [48, 49] estimates as in Eq. 2.

$$\hat{p}_{\mathrm{kg}}(x) = \hat{p}_k(Y = g|X = x) = \frac{1}{k} \sum_{i \in N_k(x,\,D)} I(y_i = g) \tag{2}$$

The K value in the K-NN is used to select the closest features. The point whose class is to be determined is assigned to the most common class by looking at its K-nearest neighbors. For each sample whose class is to be estimated, the processing load increases in case the dataset grows due to the nearest neighbor search among all samples in the dataset. Since the K-NN algorithm is distance-based, normalization of the training data can significantly improve the accuracy of the classifier. K-NN is an algorithm that produces simple and fast results, shows high performance on multidimensional data and is easy to implement.

Table 4 Statistical characteristic of the dataset

Features	Average	Standard deviation	Minimal	Maximal
aOrtalama	319,841	8,05662	301,111	346,929
agaz	328,3532	17,40,725	274	380
agaz_median	328,6052	15,85,599	290	380
agaz_ort	327,0144	15,27,127	290,5	379,6897
atepe	318,3849	15,55,081	296	371
birinciGuclu	321,9802	14,57,781	296	380
birinciGucluSure	12,97,619	4,874,512	4	32
birinciMikroGuclu	316,123	17,59,893	0	365
dif_Rtm_Val	3,543,651	0,94,484	2	7
enKisaRtm	24,25,397	11,89,212	8	64
enPest	298,625	7,307,792	243	305
enTiz	343,9345	14,34,361	318	396
enUzunRtm	3,986,111	1,504,733	1	8
ikinciGuclu	320,8075	12,73,482	296	365
ikinciGucluSure	12,00794	8,775,527	4	64
ikinciMikroGuclu	315,1468	21,15,517	0	376
ilkNota	319,4585	19,23,063	274	380
notaCarpiklik	-0,03746	0,343,314	-1,192	0,983
ortalama	12,80,648	4,262,736	5,263	37,574
ortanca	321,0188	9,257,748	296	358
rtmCarpiklik	0,432,972	0,936,035	-5,295	2,838
rtmOrtalama	12,80,648	4,262,736	5,263	37,574
sesSistemi	7,301,587	2,65,294	3	26
sonNota	304,3472	3,246,725	296	349
stdPerde	12,0081	3,777,059	5,377	30,347
toplamNota	130,2599	128,4798	16	1260
ucuncuGuclu	319,381	13,23,851	296	371
varyansPerde	158,4323	111,4482	28,912	920,94

Random forest (RF) algorithm, which is based on the method created by Ho [60] and later developed and brought to the literature by [61], is a collective learning technique that determines the output class by creating a large number of decision trees during training and taking the mode or average of the results produced by each tree. RF solves the problem of overfitting, which is one of the common problems in traditional decision trees, by dividing both the dataset and attributes into many parts and processing them on multiple trees. RF is a frequently preferred algorithm due to its high prediction performance, its ability to cope with imbalance problems, and its ability to produce reliable results in a wide range of applications.

Logistic regression (LR) is a statistical model used in binary classification problems where the dependent variable is discontinuous. Computer science is widely used in many applied science and real-world problems. Logistic regression is a predictive analysis to explain the relationship between a binary dependent variable and a set of independent variables. It uses a logit function to do this job. The probability of an event occurring is expressed by the formula in Eq. 3. The probability of the event not occurring is 1-p, and the logit function is calculated according to Eq. 4. Logistic regression generates the coefficients of a formula for estimating the logit transform.

$$p = \frac{e^{a+bx}}{1 + e^{a+bx}} \tag{3}$$

$$\mathrm{logit}(p) = \ln\left(\frac{p}{1-p}\right) \tag{4}$$

LR is an algorithm that is characterized by fast training, low computational cost and simplicity, and is widely used in computer science, many applied sciences and real-world problems.

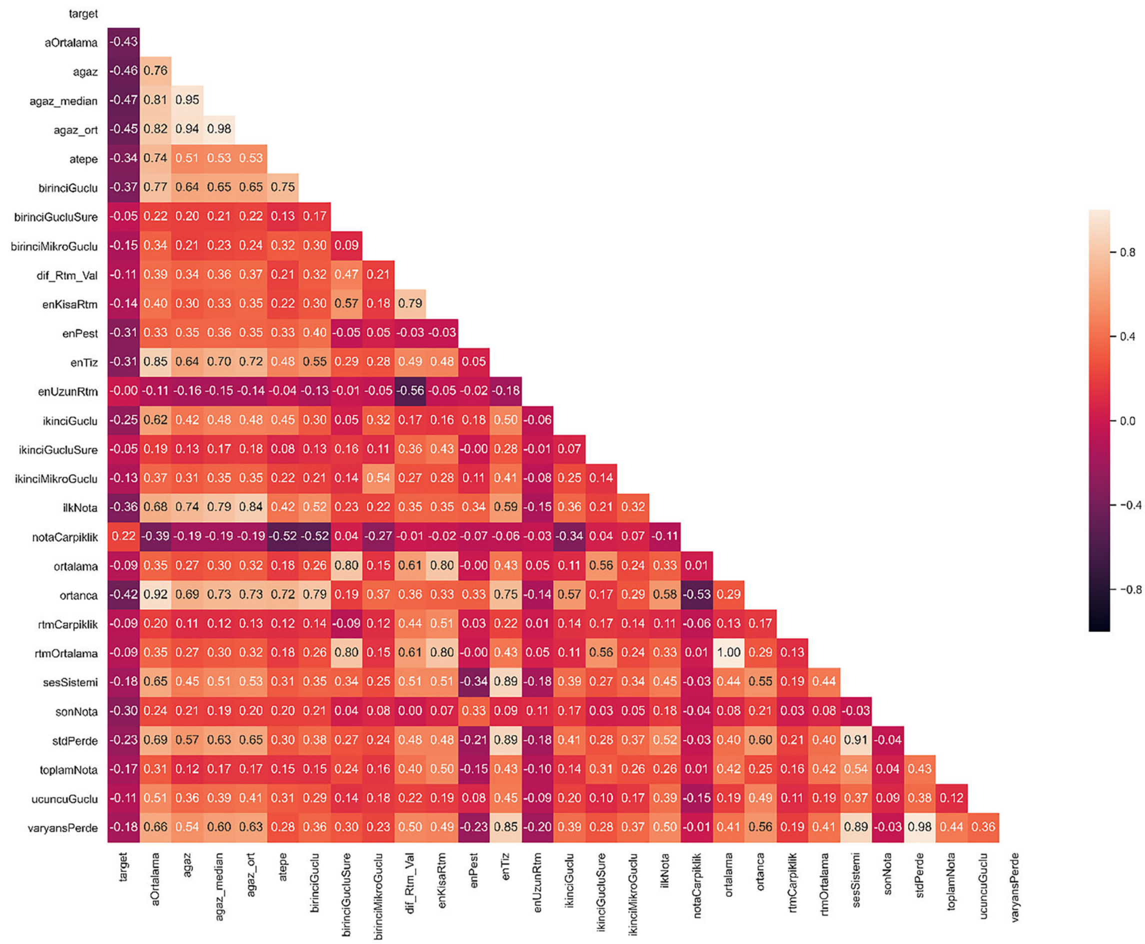

Fig. 2 Correlation matrix

3 Experimental study

The dataset used in the study consists of 504 samples defined by 28 features and 11 class values. All 28 input features are of numeric (continuous-ordered) type. The target class value is of categorical type, including Gülizar (1), Hicaz (2), Uzzal (3), Hüseyni (4), Muhayyer (5), Muhayyer-Kürdi (6), Nevruz-ı Asıl (7), Nikriz (8), Nühüft-i Kadim (9), Rast (10), Uşşak (11). Before the classification process, some pre-processes were applied to the data to increase the prediction success. In this context, since "es" values are expressed with "-1" in SymbTr file format, if the piece starts with "es," these lines are deleted in order to determine the starting pitch. The "es" values at the end of the piece have been replaced with the MIDI pitch value of the previous pitch. The majority of the TFM repertoire consists of oral pieces. Since TFM generally does not have a compositional seyir [62], the instrumental parts at the beginning of the melodies are mostly the same as the chorus parts of the pieces. On the other hand, in the pieces in the uzun hava form (pieces having no time), a compositional approach can be seen especially in the improvisation parts at the beginning of the pieces. In order to determine the makam correctly, the instrumental parts of the pieces in the dataset were excluded and not included in the research. At the same time, eight missing data in the "agaz_median," "agaz_ort," "secondMikroGuclu" and "ucuncuGuclu" columns were filled with the average value of the relevant column. All numeric input values are then scaled to the range 0–1. The distribution of the data according to the authorities is given in Fig. 3.

As seen in Fig. 3, there is a significant imbalance in the makam distributions in the dataset. At the same time, a distribution between attributes and class labels for each artifact in the dataset is presented in Fig. 4 with the t-distributed stochastic neighbor embedding (t-SNE) graph. t-SNE is a popular method for exploring structure in high-dimensional, complex datasets while preserving stronger local relationships between data points [63]. t-SNE, a nonparametric, unsupervised machine learning method, uses a stepwise iterative approach to find a lower-dimensional representation of the original data while preserving knowledge about the local neighborhood of the data. The

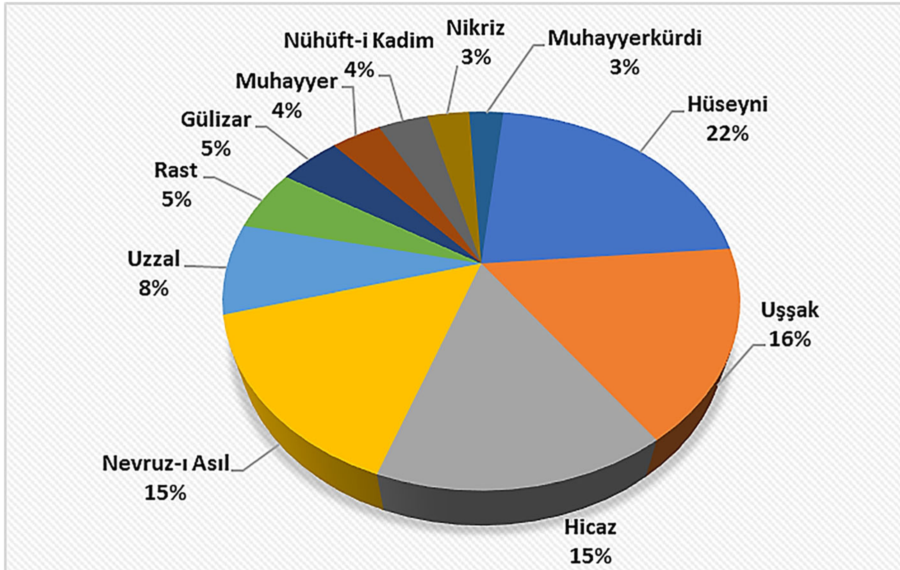

Fig. 3 Distribution of makams in the dataset

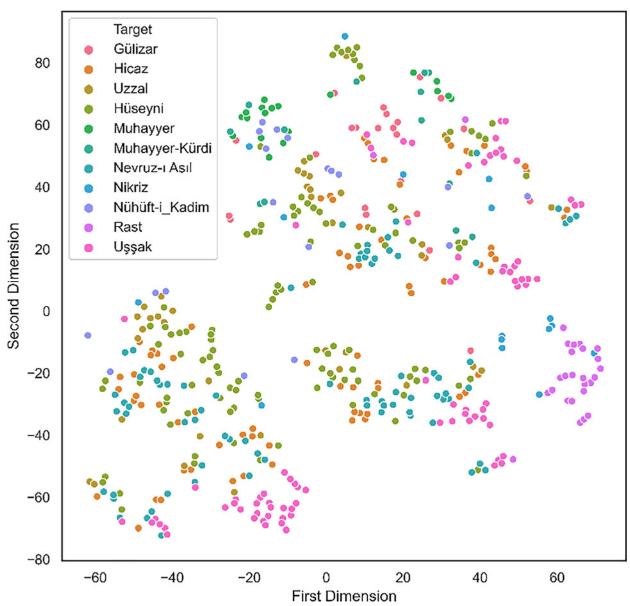

Fig. 4 t-SNE feature representation

parameters used to create the t-SNE representation and their values are: n_components = 2, perplexity = 10, learning_rate = 200, n_iter = 5000, metric = "euclidean," init = "random," method = "barnes_hut."

According to the t-SNE graph in Fig. 4, it is seen that there are overlaps at certain points in the makam distributions. This may cause difficulties in distinguishing makams. In addition, in order to see how the data on the pieces are distributed in different makam structures using the same tone/fret tunings, a distribution on the makam structures that make up 90% of the pieces is presented in Fig. 5.

In Fig. 5, it is seen that there are overlap situations especially between the makam structures using the same tone/fret tunings. The reason why the distinction between classes is not clear at certain points is due to the similarity in characteristics, as well as the class imbalances caused by the smaller number of samples in certain classes. The reason for the imbalance in the original dataset is that there are not enough pieces for each makam class in the TFM repertoire. In such cases, since the classification model used tends to learn the dominant data, resampling techniques are used to find a solution to this problem by making the dataset more balanced. Oversampling, which is a data augmentation technique by resampling, is to ensure class balance by multiplying minority class data in accordance with the statistical characteristics of the data. For this work, synthetic data generation approaches are applied by randomly selecting the scarce data and copying or interpolation methods. In this study, in addition to the default classification, four different oversampling techniques were used in order to reveal the effect on the model success, since the imbalance in the dataset used was large. These SMOTE are KMeansSMOTE, ADASYN, and SVMSMOTE. SMOTE is an approach to synthetic generation from minority data developed by [64]. KMeansSMOTE is a technique that applies K-means clustering over the SMOTE technique to reduce noise generation [65]. Similarly, the SVMSMOTE technique combines SMOTE with a SVM [66]. The ADASYN technique is an adaptive, synthetic oversampling technique that is similar to the SMOTE technique but produces a varying number of samples depending on the class distributions in the data [67]. Performance measurements were carried out by applying the above-mentioned resampling methods separately for each of the different classifiers on the dataset.

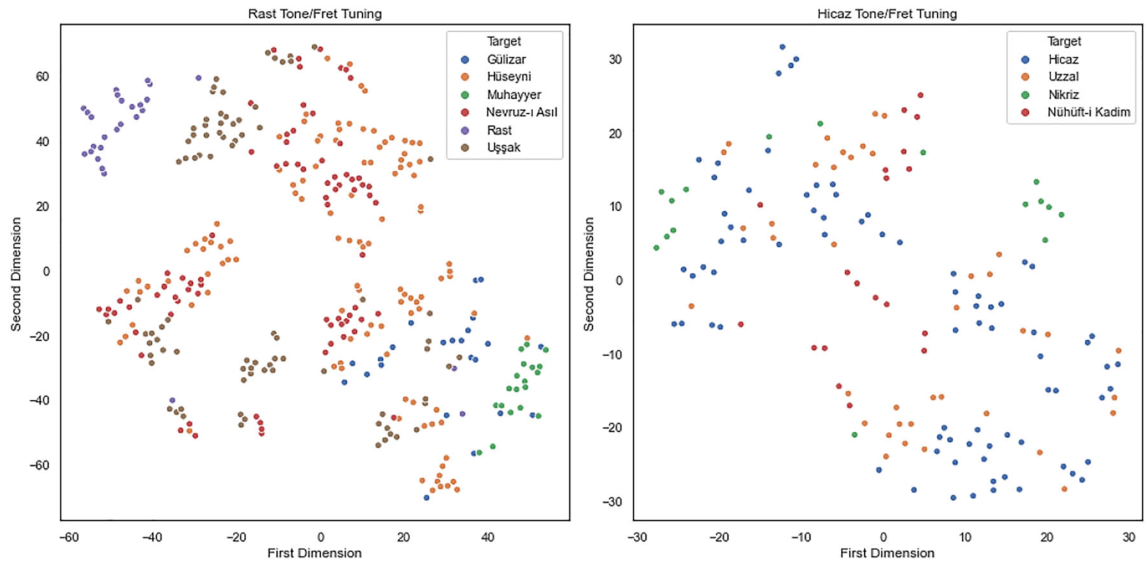

Fig. 5 t-SNE feature representation according to tone/fret tunings

In measuring the success of classifiers, metrics calculated over correlation matrices were used. Complexity matrices provide detailed information about which classes can be distinguished from each other at the end of the classification process. The metrics used in this study are presented in Table 5.

The accuracy metric, seen in Table 5, is used to express the overall success of the model. Accuracy value is obtained by dividing the number of correctly classified samples by the total number of samples. Precision is the metric that shows how many of the positively predicted values are actually positive. Recall is a metric that shows how many of the values that should be predicted positively are predicted positively. The F1-score is used to express the balance between precision and recall. The F1-score is the harmonic mean of the calculated precision and recall values. The F1-score takes values in the range of [0, 1].

4 Results and discussion

LGBM, XGB, NB, DT, SVM, K-NN, RF and LR classifiers and SMOTE, KMeansSMOTE, ADASYN, and SVMSMOTE over sampling techniques were applied for each classifier and performance measurements were performed on the dataset used in the study. The hyperparameters of the classifier were determined by the grid search method. The parameters of each classifier used for the study are given in Table 6.

Table 5 Performance metrics

Metrics	Mathematical expression
Accuracy	(TP + TN)/(TP + FP + FN + TN)
Precision	TP/(TP + FP)
Recall	TP/(TP + FN)
F1-score	2 * precision * recall/(precision + recall)

The results of each classification process with and without data augmentation with resampling are reported over the metrics given in Table 5. The data increase rate was applied to bring the minority and dominant class sample numbers closer to each other. The sample size of the dataset varies between 1015 and 1236 as a result of the data augmentation processes performed with 4 different techniques to ensure class balances. In the training phase of the models, tenfold cross-validation method was used. The averages of the results obtained at each stage are reported. Examples of the classifications made on the test data are presented separately for each classifier in Fig. 6. Measurements showing the effect of data augmentation techniques by resampling on classifiers are in Table 7 for LGBM, Table 8 for XGB, Table 9 for NB, Table 10 for DT, Table 11 for SVM, Table 12 for K-NN, Table 13 for RF and Table 14 for LR.

When the obtained measurements were evaluated, it was observed that data augmentation techniques increased the success of all classifiers compared to the without data augmentation situation. In terms of performance metrics, it has been observed that up to 29% performance increase has been achieved by making data augmentation again compared to the cases without data augmentation. Although the application of the data augmentation method by resampling has not been found in similar makam recognition studies in the literature, it supports similar studies in other fields in the literature, and it has been concluded that the increase in

Table 6 Parameters of the classifiers

Classifiers	Parameters
LGBM	Learning_rate = 0.1, n_estimators = 100, boosting_type = 'gbdt
XGB	Learning_rate = 0.1, n_estimators = 100, booster = 'gbtree'
NB	Default
DT	Criterion = "gini," splitter = "best,"
SVM	Kernel = "linear," C = 3
K-NN	n_neighbors = 3
RF	n_estimators = 200
LR	C = 0.1, penalty = "l2," max_iter = 250

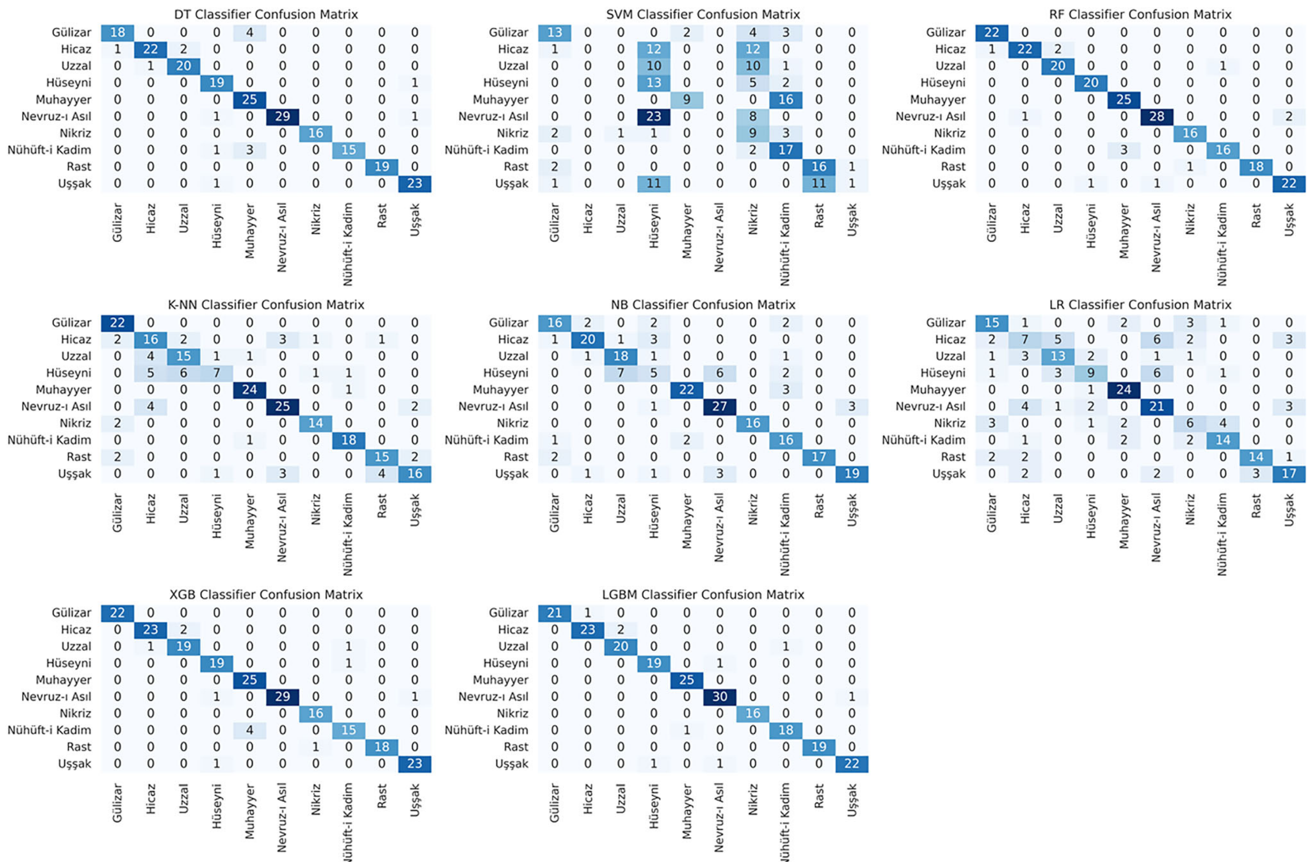

Fig. 6 Confusion matrices of the classifications made by each machine learning technique

Table 7 Classification results for light gradient boosting machine (LGBM)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.88644	0.89105	0.88644	0.87761	504
With data augmentation	SMOTE	**0.99173**	**0.99242**	**0.99173**	**0.99171**	**1221**
	KmeansSMOTE	0.96595	0.96891	0.96595	0.96530	1232
	ADASYN	0.97271	0.97404	0.97271	0.97259	1200
	SVMSMOTE	0.95890	0.96198	0.95890	0.95832	1053

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

Table 8 Classification results for eXtreme gradient boosting (XGB)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.88371	0.88431	0.88371	0.87297	504
With data augmentation	SMOTE	**0.98347**	**0.98484**	**0.98347**	**0.98343**	**1221**
	KmeansSMOTE	0.95670	0.95952	0.95670	0.95635	1236
	ADASYN	0.95849	0.96264	0.95849	0.95818	1200
	SVMSMOTE	0.95578	0.96004	0.95578	0.95478	1084

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

Table 9 Classification results for naive Bayes (NB)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.69167	0.72534	0.69167	0.68711	504
With data augmentation	SMOTE	0.77734	0.80254	0.77734	0.76857	1221
	KmeansSMOTE	**0.85732**	**0.87354**	**0.85732**	**0.85113**	**1233**
	ADASYN	0.74328	0.76643	0.74328	0.72907	1200
	SVMSMOTE	0.80282	0.83613	0.80282	0.79784	1015

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

Table 10 Classification results for decision trees (DT)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.81364	0.82214	0.81364	0.80511	504
With data augmentation	SMOTE	0.92892	0.93344	0.92892	0.92873	1221
	KmeansSMOTE	**0.93797**	**0.94124**	**0.93797**	**0.93748**	**1233**
	ADASYN	0.90829	0.91736	0.90829	0.90729	1200
	SVMSMOTE	0.91103	0.92017	0.91103	0.91052	1073

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

Table 11 Classification results for support vector machines (SVM)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.70192	0.71049	0.70192	0.68972	504
With data augmentation	SMOTE	0.92555	0.93464	0.92555	0.92306	1221
	KmeansSMOTE	0.90482	0.91254	0.90482	0.90343	1230
	ADASYN	**0.92942**	**0.93573**	**0.92942**	**0.92735**	**1200**
	SVMSMOTE	0.90208	0.90923	0.90208	0.90020	1022

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

this study is quite high. Among the applied data augmentation techniques, it was determined that the highest increase in success was achieved with SMOTE and KmeansSMOTE. The best results of all used classifiers and data augmentation techniques are summarized in Table 15.

When the results in Table 15 are examined, it is seen that the highest measurements in terms of all metrics are obtained with LGBM, which is one of the ensemble learning methods. The highest accuracy value obtained with this classifier was obtained with SMOTE, one of the data augmentation techniques. The best results obtained in all classifications with LGBM were 99.17% for accuracy, 99.24% for precision, 99.17% for precision, and 99.17% for F1-score. After LGBM, the highest success rate was achieved with XGB. The highest accuracy value of 98.34% reached with this classifier was also obtained with the

Table 12 Classification results for K-nearest neighbor (K-nn)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.57833	0.58356	0.57833	0.56357	504
With data augmentation	SMOTE	**0.88154**	**0.89613**	**0.88154**	**0.87448**	**1221**
	KmeansSMOTE	0.85841	0.87116	0.85841	0.85329	1233
	ADASYN	0.85431	0.86980	0.85431	0.84810	1200
	SVMSMOTE	0.84545	0.85985	0.84545	0.84009	1071

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

Table 13 Classification results for random forest (RF)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.85860	0.86335	0.85860	0.84988	504
With data augmentation	SMOTE	**0.96494**	**0.96721**	**0.96494**	**0.96441**	**1221**
	KmeansSMOTE	0.95549	0.95972	0.95549	0.95481	1233
	ADASYN	0.95350	0.96052	0.95350	0.95236	1200
	SVMSMOTE	0.95121	0.95659	0.95121	0.95042	1063

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

Table 14 Classification results for logistic regression (LR)

Data augmentation	Method	Accuracy	Precision	Recall	F1-score	Number of samples (N)
Without data augmentation	–	0.58922	0.54869	0.58922	0.55601	504
With data augmentation	SMOTE	0.78519	0.79052	0.78519	0.77777	1221
	KmeansSMOTE	**0.81485**	**0.83019**	**0.81485**	**0.80849**	**1232**
	ADASYN	0.73926	0.75361	0.73926	0.72646	1200
	SVMSMOTE	0.77537	0.78486	0.77537	0.77022	1021

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

SMOTE technique. In this study, the lowest accuracy values among the classifiers tested using data augmentation techniques belong to the logistic regression classifier with 81.48%. The best precision values for all classifiers range from 83.01 to 99.24%. The best recall values are between 81.48 and 99.17%. This shows that all classifiers have a high ability to detect true positives and eliminate false positives. The best results obtained in this study and the results of recent similar studies in the literature are given in Table 16 in comparison.

In this study, it was seen that the classifiers used with data augmentation through applied resampling showed higher success than the general of similar studies in the literature. It is seen that the LGBM model used in this study, together with data augmentation, provides higher accuracy with 99.17% accuracy than all similar studies in Table 16 in recent years. The highest accuracy value obtained in 15 similar studies in Table 16 is 98.21%, and the lowest accuracy is 70%. The average accuracy value reached in the studies in the literature is 88.65%. The lowest accuracy value obtained in this study is 81.48%, and the average accuracy of the eight proposed models is 91.26%. In general, the accuracy values obtained with the models applied in this study are higher than the accuracy values obtained in other studies in the literature.

In the studies in the literature, it is seen that the highest accuracy is obtained with the SVM model and various neural network models. In this study, the highest accuracy value was obtained by using the data augmentation approach together with the boosting methods such as LGBM and then XGB. Boosting algorithms are inherently less resistant to overfitting problems. It is seen that SMOTE and KmeansSMOTE methods, which are data augmentation techniques by resampling, are more successful than other similar techniques. Achieving class equilibria through data augmentation also reduces susceptibility to overfitting a particular class. In this sense, it can be said based on the results obtained that the combination of

Table 15 The best results obtained related to data augmentation

Classifier	Method	The best accuracy	The best precision	The best recall	The best F1-score
LGBM	Without data augmentation	0.88644	0.89105	0.88644	0.87761
	With data augmentation	**0.99173**	**0.99242**	**0.99173**	**0.99171**
XGB	Without data augmentation	0.88371	0.88431	0.88371	0.87297
	With data augmentation	**0.98347**	**0.98484**	**0.98347**	**0.98343**
NB	Without data augmentation	0.69167	0.72534	0.69167	0.68711
	With data augmentation	**0.85732**	**0.87354**	**0.85732**	**0.85113**
DT	Without data augmentation	0.81364	0.82214	0.81364	0.80511
	With data augmentation	**0.93797**	**0.94124**	**0.93797**	**0.93748**
SVM	Without data augmentation	0.70192	0.71049	0.70192	0.68972
	With data augmentation	**0.92942**	**0.93573**	**0.92942**	**0.92735**
K-NN	Without data augmentation	0.57833	0.58356	0.57833	0.56357
	With data augmentation	**0.88154**	**0.89613**	**0.88154**	**0.87448**
RF	Without data augmentation	0.85860	0.86335	0.85860	0.84988
	With data augmentation	**0.96494**	**0.96721**	**0.96494**	**0.96441**
LG	Without data augmentation	0.58922	0.54869	0.58922	0.55601
	With data augmentation	**0.81485**	**0.83019**	**0.81485**	**0.80849**

Bold values indicate the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

resampling and data augmentation with ensemble learning and boosting has a stronger effect on the solution of the overfitting problem and thus increases the generalizable success of the model. Examples of the classifications made by each machine learning technique are presented in Fig. 6 to demonstrate its effectiveness.

When Fig. 6 is analyzed, it can be seen that there are significant differences in the classifiers' discrimination levels of certain classes. The majority of the classifiers seem to be quite successful in distinguishing modes such as Gülizar, Uşşak, Rast, Nühüft-i Kadim. However, especially SVM and LR classifiers tended to discriminate the data less than the other classifiers. In general, the classifiers discriminated the class instances at a high rate.

Furthermore, the limitations and scope of this study should be taken into account. The available dataset is based on a limited set of artifacts, consisting of only 504 samples. This limitation is a consequence of the accessibility of the available artifacts and the high cost of the digitization processes. In particular, the resources and time required to digitize the artifacts have limited the size of the dataset. The attributes on which the classification method used in this study is based have been carefully selected to meet the aims and objectives of this study. However, in order to better understand the general validity of these features, it is thought that they should be tested in different studies with different analysis methods and feature selection techniques. This may provide a solid basis for the transferability of the results of the study to a more general context and their validity under various conditions. In conclusion, the limitations of the study are shaped by factors such as the size of the available dataset and the specificity of the features used for the specific purposes of the study. Being aware of these points are important considerations when interpreting the results of the study and designing similar studies in the future.

5 Conclusions and recommendations

In this study, an alternative solution has been proposed by using different machine learning algorithms for the problem of makam recognition in TFM, which has been the subject of various perspectives and discussions for a long time. Determining the makams of TFM pieces can create a difficult situation, especially since the works do not show a compositional seyir. The piece is mostly sung impromptu or created instantaneously, and it often emerges as a situation in which the patterns that exist in the musical memory of the person are revealed as they come from within. The instantaneous combination of the patterns existing in the infinite seyir space by the performer can cause various difficulties in revealing the makam phenomenon, which includes a compositional seyir feature. However, this does not mean that the work is far from any makam belonging. It is clear that solving such a complex phenomenon with machine learning approaches will make positive contributions to music information access. In this study, a unique dataset consisting of 504 examples with 28 different input parameters for recognizing makams specific

Table 16 Comparison of the findings with the studies in the literature

References	Methods	Dataset	The best accuracy (%)
Kalaycı and Korukoğlu, 2012 [42]	K-means, ANN	Bir Şarkıdır Yaşamak—Turkish Art Music Selection Series (103 pieces, 7 makams)	70.00
Ioannidis et al., 2011 [40]	Chroma features	Recordings from commercial CDs (302 pieces, 9 makams)	74.17
Gedik and Bozkurt, 2010 [39]	Pitch frequency histogram	Various recordings (172 pieces, 9 makams)	77.38
Ünal et al., 2014 [20]	A hierarchical approach	SymbTr (877 pieces, 13 makams)	87.90
Öztürk et al., 2018 [18]	J48, MLP, random forest, logistic, bagging, SMO, naive Bayes, BayesNet, JRIP, LMT, N-Gram	SymbTr (1246 pieces, 13 makams)	88.12
Demirel et al., 2018 [46]	Chroma features	The Ottoman-Turkish makam recognition dataset (997 pieces, 20 makams)	89.00
Kızrak et al., 2014 [43]	Probabilistic neural networks	Recordings from commercial CDs and Internet (62 pieces, 6 makams)	89.40
Abidin et al., 2017 [13]	Karar Ağacı	SymbTr (1261 pieces, 13 makams)	89.70
Börekci, 2022 [48]	XGBoost, Bagging, K-NN, Logistic Regression, SVM, Naive Bayes	Turkish Folk Music Repertoire/SymbTr (437 pieces, 8 makams)	90.15
Gedik and Bozkurt, 2008 [38]	Pitch interval histogram	Synthetic and real audio files (118 pieces, 9 makams)	92.00
Kızrak and Bolat, 2015 [44]	Deep belief networks	Recordings from commercial CDs and Internet (62 pieces, 6 makams)	92.70
Kalender et al., 2012 [41]	Combined neural networks	Selçuk University Dilek Sabancı State Conservatory Archives (50 pieces, 9 makams)	95.83
Kızrak and Bolat, 2017 [45]	Deep belief networks	Recordings from commercial CDs and the Internet (93 pieces, 7 makams)	96.57
Alpkoçak and Gedik, 2005 [37]	N-gram	MIDI files (200 pieces, 10 makams)	97.7
Kayış et al., 2021 [47]	Support vector machines	Audio files (8 makams, 448 pieces)	98.21
Literature average			88.65
Recommended LGBM			**99.17**
Recommended XGB			98.34
Recommended NB			85.73
Recommended DT			93.79
Recommended SVM			92.94
Recommended K-NN			82.15
Recommended RF			96.49
Recommended LR			81.48
Average of recommended models			91,26

Bold value indicates the highest performance rate for each classification model, in particular the highest performance rates obtained from data augmentation techniques

to this study was studied. Classifications for makam recognition were carried out by using 8 different machine learning algorithms and data augmentation techniques through four different resampling to increase the prediction accuracy by balancing the dataset. According to the results of 160 measurements characterized by 40 different classification accuracy, precision, recall, and F1-score, the highest accuracy of 99.17% was obtained when the light gradient boosting machine classifier was used together with the SMOTE data augmentation technique. The majority of the methods proposed in the study showed higher success than the studies in the literature. The highest accuracy value obtained is higher than all the studies in the literature. The main factor that determines the success of machine learning algorithms that learn from data is the dataset studied. Class imbalances in the dataset negatively affect the success of the model and the generalizability of the results. Balancing the available dataset by using appropriate sampling techniques allows the used model to make more successful classifications. The results obtained in this study formed an opinion about the performance of resampling and data augmentation techniques and allowed comparing the success of different machine learning models. In the classifications made on the unbalanced dataset used in this study, it was seen that data augmentation increased success up to 29%. In addition, an original dataset has been brought to the field for future research. According to the research results, some suggestions for potential future research are as follows: The use of a larger and more diverse dataset in future research may improve the generalization capabilities of makam recognition algorithms. A dataset that includes samples from different periods, regions and performers may better reflect the algorithm performance. Further research on the effect of the attributes used may increase the likelihood of better results. Experimenting with different feature sets and evaluating their impact on authority recognition accuracy may better guide the choice of features for future studies. Experimental evaluation of different machine learning algorithms is important. Model selection and hyperparameter tuning can be performed over a wider range of domains to focus on achieving the best performance. The effect of different sampling and boosting techniques on the data imbalance problem can be analyzed in more depth. This could significantly improve the overall performance of authority recognition algorithms. The optimization of makam recognition algorithms for real-time applications and their usability in mobile applications or digital music platforms can be investigated.

Acknowledgements In this study, the new version of the dataset previously created by Börekci[48], restructured for this study, updated and expanded was used. In this context, in the first study, the dataset consisting of 21 features and 437 artifacts was expanded to a total of 28 features and 504 sample artifacts, with 7 new features and 67 new samples.

Data availability The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

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